

Haithem Turki

Curriculum Vitae

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EDUCATION

Carnegie Mellon University Ph.D. in Computer Science Thesis: Towards City-Scale Neural Rendering Advisor: Deva Ramanan (committee chair) Committee: Jessica K. Hodgins, Martial Hebert, Jonathan T. Barron	2019 - 2024
Stanford University M.S. in Computer Science	2012 - 2018
Stanford University B.S. in Computer Science, minor in Management Science and Engineering	2008 - 2012

EXPERIENCE

NVIDIA , Senior Research Scientist Spatial Intelligence Lab with Zan Gojcic and Sanja Fidler	2024-
Meta , Research Scientist Intern w/ Christian Richardt, Michael Zollhöfer	2023
Argo AI , Research Intern w/ Deva Ramanan, Francesco Ferroni	2022
Page One Ventures , Venture Partner	2020 - 2024
Palantir Technologies , Engineering Manager and Technical Lead	2012 - 2019
Addepar , Engineering Intern	2011
Stanford University , Research Assistant w/ Daphne Koller, M. Pawan Kumar	2010

AWARDS & HONORS

Best Paper Award Candidate , CVPR 2025 DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models	2025
Highlight Paper , CVPR 2024 (top 2.8% of submissions) HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces	2024
Highlight Paper , CVPR 2024 (top 2.8% of submissions) SpecNeRF: Gaussian Directional Encoding for Specular Reflections	2024

PUBLICATIONS

- [18] Alessandro Burzio*, Tobias Fischer*, Sven Elflein, Qunjie Zhou, Riccardo de Lutio, Jiawei Ren, Jiahui Huang, Shengyu Huang, Marc Pollefeys, Laura Leal-Taixé, Zan Gojcic†, **Haithem Turki†**, “Déjà View: Looping Transformers for Multi-View 3D Reconstruction”, Preprint

- [17] Riccardo de Lutio*, Tobias Fischer, Yen-Yu Chang, Yuxuan Zhang, Jay Zhangjie Wu, Xuanchi Ren, Tianchang Shen, Katarina Tothova, Zan Gojcic, **Haithem Turki***, “ArtiFixer: Enhancing and Extending 3D Reconstruction with Auto-Regressive Diffusion Models” in ACM SIGGRAPH, 2026, **(Invited Oral, GenRecon3D Workshop @ CVPR 2026)**
- [16] Yuxuan Zhang*, Katarina Tothova*, Zian Wang, Kangxue Yin, **Haithem Turki**, Riccardo de Lutio, Yen-Yu Chang, Or Litany, Sanja Fidler, Zan Gojcic, “DiffusionHarmonizer: Bridging Neural Reconstruction and Photorealistic Simulation with Online Diffusion Enhancer” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [15] **Haithem Turki***, Qi Wu*, Xin Kang, Janick Martinez Esturo, Shengyu Huang, Ruilong Li, Zan Gojcic, Riccardo de Lutio, “SimULi: Real-Time LiDAR and Camera Simulation with Unscented Transforms” in International Conference on Learning Representations (ICLR), 2026
- [14] Sherwin Bahmani, Tianchang Shen, Jiawei Ren, Jiahui Huang, Yifeng Jiang, **Haithem Turki**, Andrea Tagliasacchi, David B. Lindell, Zan Gojcic, Sanja Fidler, Huan Ling, Jun Gao*, Xuanchi Ren*, “Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation” in International Conference on Learning Representations (ICLR), 2026
- [13] Zhangjie Wu*, Yuxuan Zhang*, **Haithem Turki**, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic†, Huan Ling†, “DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025, **(Best Paper Award Candidate)**
- [12] **Haithem Turki**, Vasu Agrawal, Samuel Rota Bulò, Lorenzo Porzi, Peter Kotschieder, Deva Ramanan, Michael Zollhöfer, Christian Richardt, “HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, **(Highlight)**
- [11] Li Ma, Vasu Agrawal, **Haithem Turki**, Changil Kim, Chen Gao, Pedro Sander, Michael Zollhöfer, Christian Richardt, “SpecNeRF: Gaussian Directional Encoding for Specular Reflections” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, **(Highlight)**
- [10] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan, “PyNeRF: Pyramidal Neural Radiance Fields” in Conference on Neural Information Processing Systems (NeurIPS), 2023
- [9] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, “Low-Bandwidth Self-Improving Transmission of Rare Training Data” in Annual International Conference on Mobile Computing And Networking (MobiCom), 2023
- [8] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, “Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data” in International Workshop on Networked AI Systems (NetAISys), 2023
- [7] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan, “SUDS: Scalable Urban Dynamic Scenes” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- [6] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai, “Accelerating Silent Witness Storage” in IEEE Micro, Nov.-Dec. 2022

- [5] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan, “Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
- [4] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan, “OpenRTiST: End-to-End Benchmarking for Edge Computing” in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, Oct.-Dec. 2020
- [3] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng, “Edge Computing for Legacy Applications” in IEEE Pervasive Computing, Oct.-Dec. 2020
- [2] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, “Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation” in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, July 2015
- [1] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, “Learning Specific-Class Segmentation from Diverse Data” in International Conference on Computer Vision (ICCV), 2011

TECHNICAL REPORTS

- [2] Tianshi Cao*, Jiawei Ren*, Yuxuan Zhang, Jaewoo Seo, Jiahui Huang, Shikhar Solanki, Haotian Zhang, Mingfei Guo, **Haithem Turki**, Muxingzi Li, Yue Zhu, Sipeng Zhang, Zan Gojcic, Sanja Fidler, Kangxue Yin, “Asset Harvester: Extracting 3D Assets from Autonomous Driving Logs for Simulation”, arXiv:2604.18468, 2026
- [1] Ziqiang Feng, Shilpa George, **Haithem Turki**, Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan, “Improving Edge Elasticity via Decode Offload”, Technical Report CMU-CS-21-139, Carnegie Mellon University, 2021

PATENTS

- [2] **Haithem Turki**, Robert Fink, Amr Al Mallah, “Systems and Methods for Indexing and Searching.” US Patent 2019/0347343 A1, 2019
- [1] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara, “Systems and Methods for Constraint Driven Database Searching.” US Patent 2018/293239 A1, 2018

TEACHING

15-640 Distributed Systems , Carnegie Mellon University (Graduate Student Instructor)	2020 - 2021
17-700 Data Science and Machine Learning at Scale , Carnegie Mellon University (Graduate Student Instructor)	2020 - 2021
15-821 Mobile and Pervasive Computing , Carnegie Mellon University (Project Mentor)	2020 - 2021

STUDENT COLLABORATORS & INTERNS

Tobias Fischer, PhD Student @ ETH Zürich. See Publications [17, 18].

Alessandro Burzio, PhD Student @ University of Modena and Reggio Emilia. See Publications [18].

Yen-Yu Chang, PhD Student @ Cornell University. See Publications [16, 17].

INVITED TALKS

“Towards City-Scale Neural Rendering”

Stanford Vision Lab	2024
Waymo Research (virtual)	2024
Google (virtual)	2024
NVIDIA Toronto AI Lab (virtual)	2024

“Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs”

Argo AI	2022
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ACADEMIC SERVICE

Area Chair:

ACCV

Reviewer:

CVPR, ACCV, ECCV, ICCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, AAI, Transactions on Image Processing

ADDITIONAL INFORMATION

Languages:

English (native), French (fluent), Arabic (fluent), Japanese (elementary), German (elementary)

Other Qualifications:

Alchemist Accelerator (startup mentor and investor). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.